

Firas Al Haddad

+1 (514) 699-7445 | Firas.haddad.h@gmail.com | [LinkedIn](#) | [Portfolio](#) | Bilingual: English & French

SKILLS SUMMARY

Programming Languages | Python • Java • TypeScript/JavaScript • SQL • C++

Backend & Data | REST APIs • Spring Boot • PySpark • Pandas • PostgreSQL • ETL Pipelines • Data Modeling • Parquet

Cloud & Tools | Databricks • Azure DevOps • Git/GitHub • Vercel • Power BI

Frontend | React • React Native • Firebase • Tailwind CSS

EDUCATION

B.Eng. Software Engineering Co-op

Concordia University, GPA: 3.47

Expected May 2027

Montreal, QC

WORK EXPERIENCE

Data Analytics Intern

Pratt & Whitney Canada

May 2026 – Present

Longueuil, QC

- Built Python medallion pipelines centralizing 10+ finance, supplier, parts, and commodity datasets into reusable tables.
- Integrated external market-data APIs with metals sources, standardizing prices, units, categories, and mappings.
- Automated ingestion, validation, deduplication, and Parquet publishing across a five-layer data pipeline architecture.
- Built Power BI models analyzing 10 years of metals pricing, alloy costs, cost drivers, and historical trends.

Software Engineering Intern

Fonex Data Systems

May 2025 – Aug 2025

Montreal, QC

- Developed a Python BLE backend using bleak and asyncio for optical transceiver communication.
- Implemented a GATT layer to read and write SFP/QSFP EEPROM pages over Bluetooth.
- Built a multithreaded connection manager linking asynchronous device communication with a Tkinter desktop interface.

Data Analytics Intern

Pratt & Whitney Canada

Jan 2025 – Apr 2025

Longueuil, QC

- Refactored Databricks pipelines into modular PySpark workflows with reusable transformation functions.
- Added automated tests and data quality checks to improve maintainability and reduce manual validation.
- Consolidated 20+ raw data sources into analytics-ready datasets for Power BI reporting.
- Re-engineered dashboards to deliver clearer week-over-week performance insights for business users.

PROJECTS

Campus Guide | React Native, TypeScript, Google Maps API, Firebase

Concordia, QC

- Built a mobile navigation app for campus routing, indoor maps, room search, and accessible paths.
- Implemented indoor shortest-path routing using floor plans, room localization, stairs, elevators, and washrooms.
- Integrated indoor/outdoor directions with campus routing, floor selection, and room-level navigation.
- Added next-class navigation using calendar-based class detection and map-based route generation.

Mindfulness & Focus Tracker AI | React, TensorFlow.js, WebGL

Concordia, QC

- Built a real-time attention detector tracking gaze, blinks, and head pose using TensorFlow.js.
- Developed an adaptive detection pipeline with calibration, custom thresholds, and feedback triggers.
- Optimized browser inference using WebGL acceleration to reduce delay and improve smoothness.

CERTIFICATIONS

AWS Certified Data Engineer – Associate

Planned, Expected Feb. 2027